

Abdul Samad Zeeshan

samad.z@outlook.com | [in/samad-zeeshan](#) | [github.com/samad-zeeshan](#) | [Portfolio](#)

PROFILE

Computing Science graduate (2026) who builds backend services and AI features, and tests how they fail. Three internship summers building AI tools for clients.

EDUCATION

University of Alberta

Bachelor of Science, Computing Science

Sep. 2021 – Jun. 2026

- **Relevant Coursework:** Software Engineering, Databases, Data Structures & Algorithms, Web Applications & Architecture, Machine Learning, Natural Language Processing

EXPERIENCE

iTechArabian

Software Engineering Intern

Summers 2023, 2024 & 2025

- Deployed AI chatbots that answer customer questions from each client's own help content, handling about 40% of routine inquiries without staff.
- Developed an AI forecast of hotel supply orders, cutting error about 15% against manual estimates.
- Worked with a product manager, testers and client stakeholders to turn requirements into peer-reviewed features.

PROJECTS

Tally | *a small bank that never loses a cent (Java, PostgreSQL, Kubernetes, TLA+)*

[GitHub](#) | [Demo](#) | 2026

- Designed a banking API that records each transfer as two matching entries, with PostgreSQL enforcing the rules against duplicates and overdrafts.
- Crashed servers and cut the network across a distributed cluster during over 11,000 test transfers, and no money was lost or doubled.
- Answered nearly all payment requests in under 80 ms during a load test on one machine.
- Added fraud checks on how money moves between accounts, nearly doubling the fraud caught on patterns the checks had never seen.

Warden | *reviews changes to live software before they ship (Python, LangGraph, MCP, OAuth)*

[GitHub](#) | [Demo](#) | 2026

- Built an automatic reviewer that approves safe changes to live software, blocks risky ones, and asks a person when unsure.
- Stopped all 84 planted attacks that tried to trick its AI agent into approving changes.
- Limited each AI agent to one customer and one request, and blocked every forged login tried.

Docket | *reads and files receipts automatically (TypeScript, AWS, SQS, DynamoDB)*

[GitHub](#) | [Demo](#) | 2026

- Built an event-driven pipeline on AWS that reads receipts, removes personal details, and rejects any result that fails a format check.
- Keeps receipt-reading accuracy over 99.8% on test receipts, and blocks any code update that falls below a set quality bar.
- Cut the gap between the model's confidence and accuracy on receipt subtotals by over four fifths.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL, JavaScript

Backend & Cloud: REST APIs, PostgreSQL, Docker, Kubernetes, AWS (Lambda, SQS, DynamoDB, S3, CDK), FastAPI, Node.js, React, Linux

Testing & Practices: Unit, integration, load & fault testing, JUnit, pytest, CI (GitHub Actions), monitoring & alerts, code review (including AI-generated code), Agile, Git

AI & Data: LLM apps, AI agents (LangGraph, MCP), RAG, prompt engineering, model evaluation, fine-tuning (LoRA), vector search, PySpark, data modelling & warehousing (dbt, DuckDB)